Design Status

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Design Summary

PlantPal is an AI-assisted plant care system for novice and busy plant owners. Instead of a reactive Q&A helper, PlantPal centers on an AI-supported care calendar that helps users create, adjust, and follow through on routines over time. Our current design focuses on two core flows: an initial conversational setup that generates a personalized default schedule, and a temporary adjustment flow that responds to disruptions and missed tasks without permanently changing care frequency. We are exploring Human–AI interaction through mixed-initiative planning and user control, using optional explanations, calendar previews, and completion-focused reminders with snoozing and check-ins.

Scenario 1

Task and Story

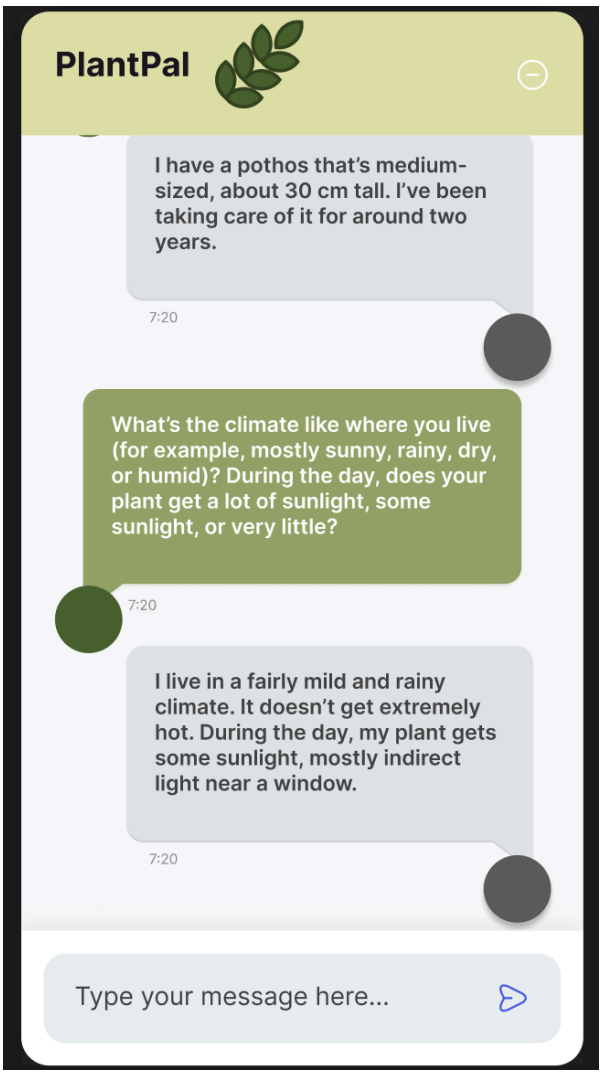
Configure a plant care schedule and sync to the user's Google Calendar

A student has an existing plant at home but wants a simple routine they can follow. In PlantPal's setup chat, the system asks a few practical questions about the plant's placement, light exposure, indoor conditions, and the user's preferences. The AI offers a clear default plan first, allowing beginners to finish quickly and customize later if needed. After the user connects Google Calendar, PlantPal generates a care schedule and syncs it into the user's calendar. Each care task includes timing guidance rather than relying on exact reminder times.

Walkthrough

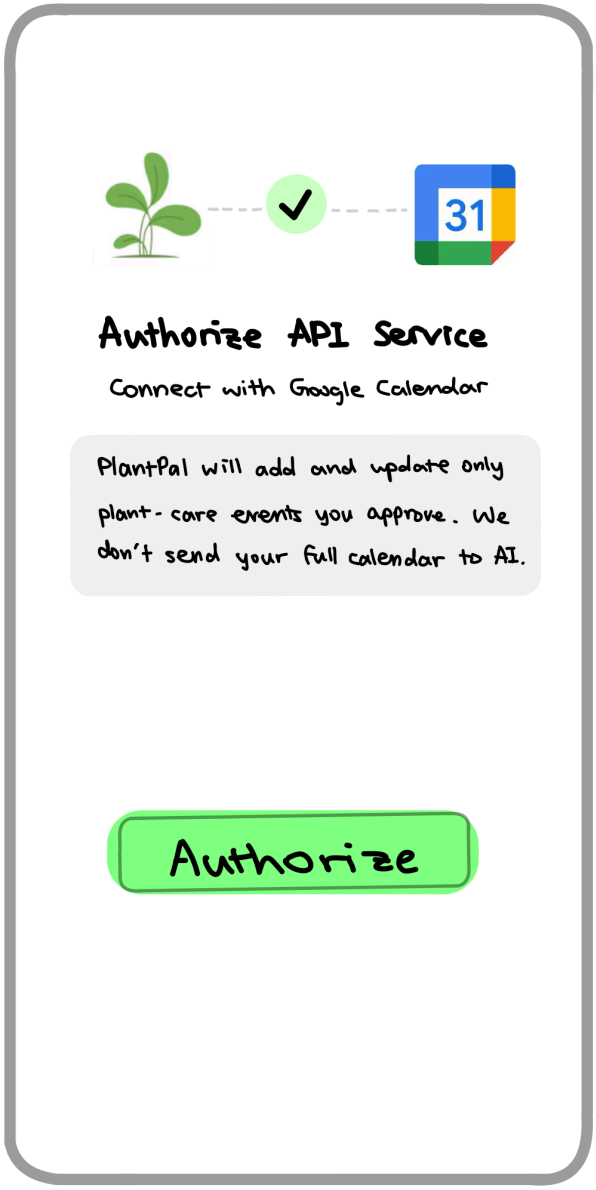
The user opens PlantPal to set up care for an existing plant and starts the setup flow from the home screen.

Key changes: setup is designed to be default-first (not form-heavy) so beginners can finish quickly.



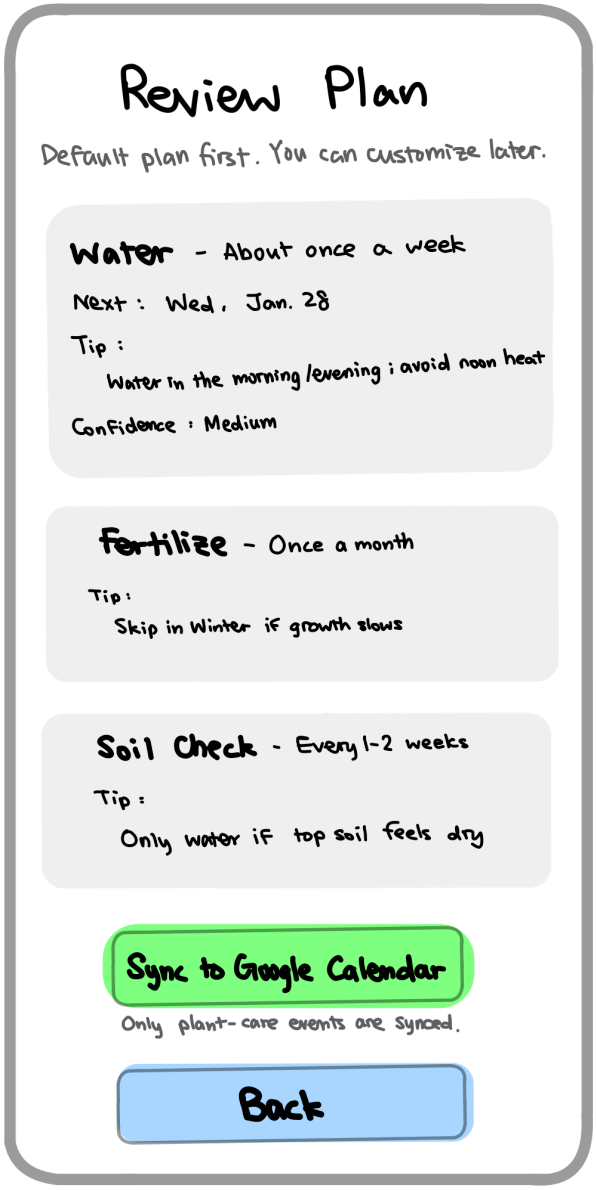
The user interacts with PlantPal through a chatbot that asks conversational questions about their location, living environment, and plant conditions. The AI leads the interaction by guiding the conversation and inferring relevant details such as climate and sunlight from the user's responses.

Key change: We replaced the old form-style setup with a conversational flow.



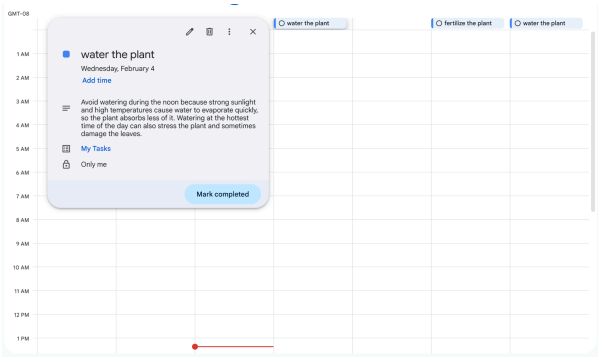
PlantPal asks the user to connect Google Calendar so the care plan can live on their normal schedule.

Key changes: we moved from an in-app calendar to Google Calendar sync via OAuth, and we clarify that only plant-care events are synced.



PlantPal generates a default care plan and shows it as a preview so the user can quickly check it before syncing.

Key changes: the plan emphasizes timing guidance and rules rather than exact reminder times.



After the user applies the plan, care tasks are added to Google Calendar and can be marked complete, supporting follow-through over time.

Key changes: each care task supports completion tracking (checkbox), which later enables reminders, snooze, and catch-up adjustments.

Language Model Support

PlantPal invokes the language model during setup to guide a short conversation and turn the user's descriptions into a workable care schedule. The model asks follow-up questions only when needed, allowing beginners to receive a default plan with minimal effort and adjust later if desired. It outputs structured care tasks with recurrence and timing guidance, along with short notes that can appear in calendar events. If the user is unsure about the plant type, the model can use a photo or description to choose a reasonable beginner-friendly plan. Only plant-care data is shared with the model.

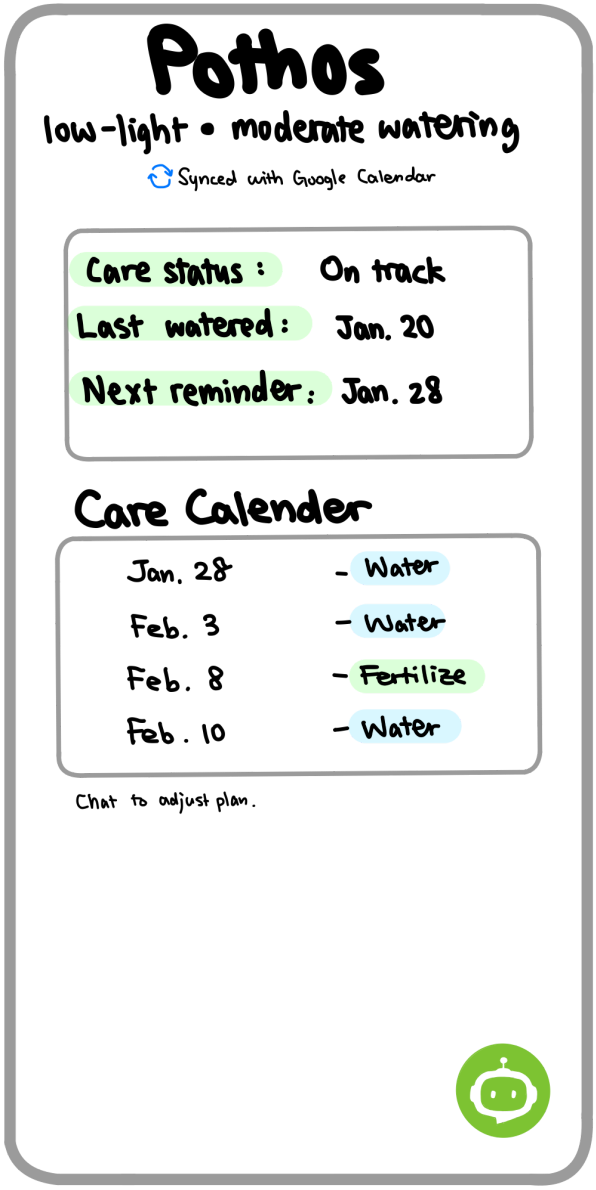
Scenario 2

Task and Story

Adjust an existing PlantPal care schedule

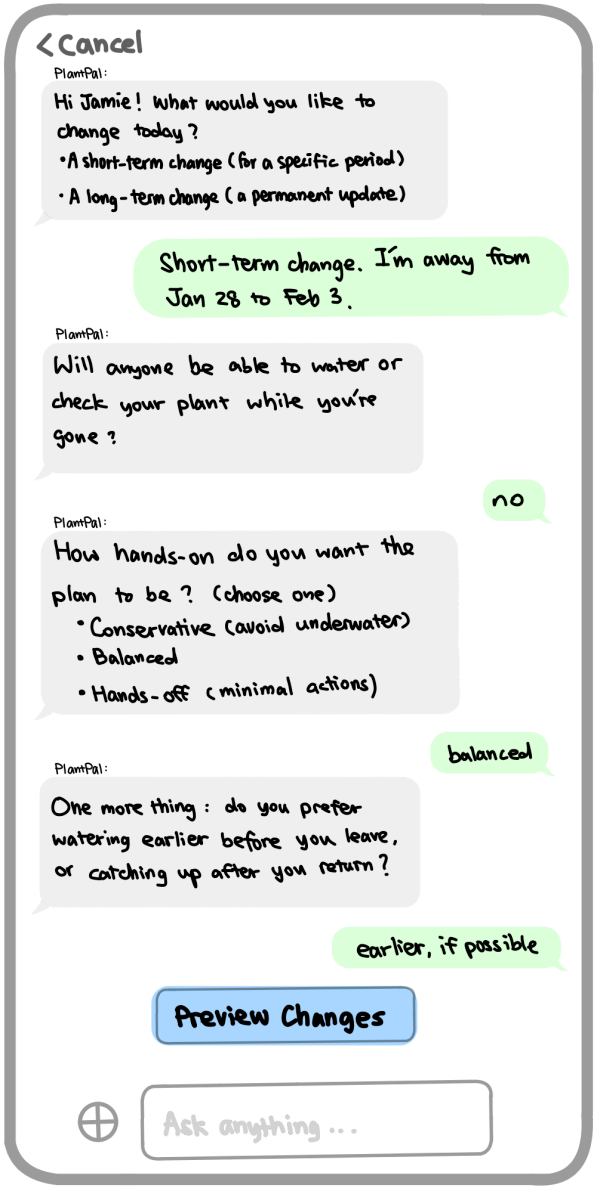
Jamie realizes they'll be away Jan 28–Feb 3 and doesn't want their plant to miss watering. They open PlantPal and describe the situation in chat. PlantPal asks quick follow-ups (dates, whether anyone can help, how strict the plan should be), then shows a preview of proposed changes: a one-time early watering, pausing reminders during travel, and resuming the regular schedule after return. Jamie reviews the preview, optionally tweaks the day, and applies the plan so it appears in Google Calendar. On the next watering day, PlantPal checks completion and supports snoozing.

Walkthrough



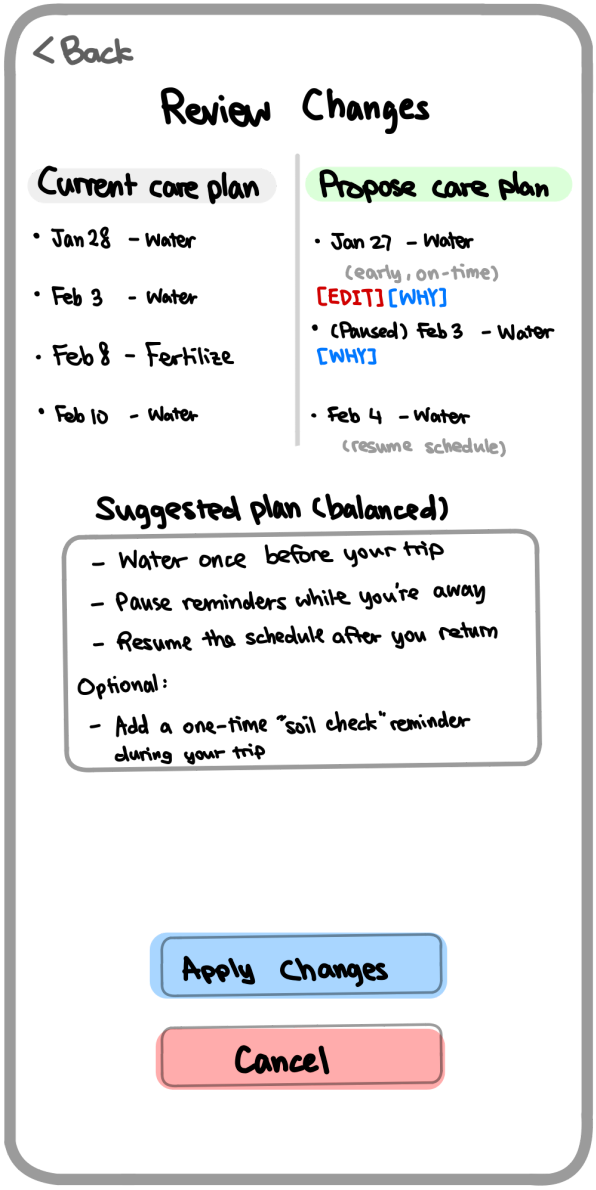
Jamie opens PlantPal and lands on the plant home screen. The app shows a quick status summary (Care status, Last watered, Next reminder) plus an upcoming Care Calendar list. Jamie can tap the chat button to adjust the plan.

Key changes: the calendar is framed as a simple reminder list (not time-precise scheduling), and adjustments start through chat, not a fixed “I’ll be away” menu option.

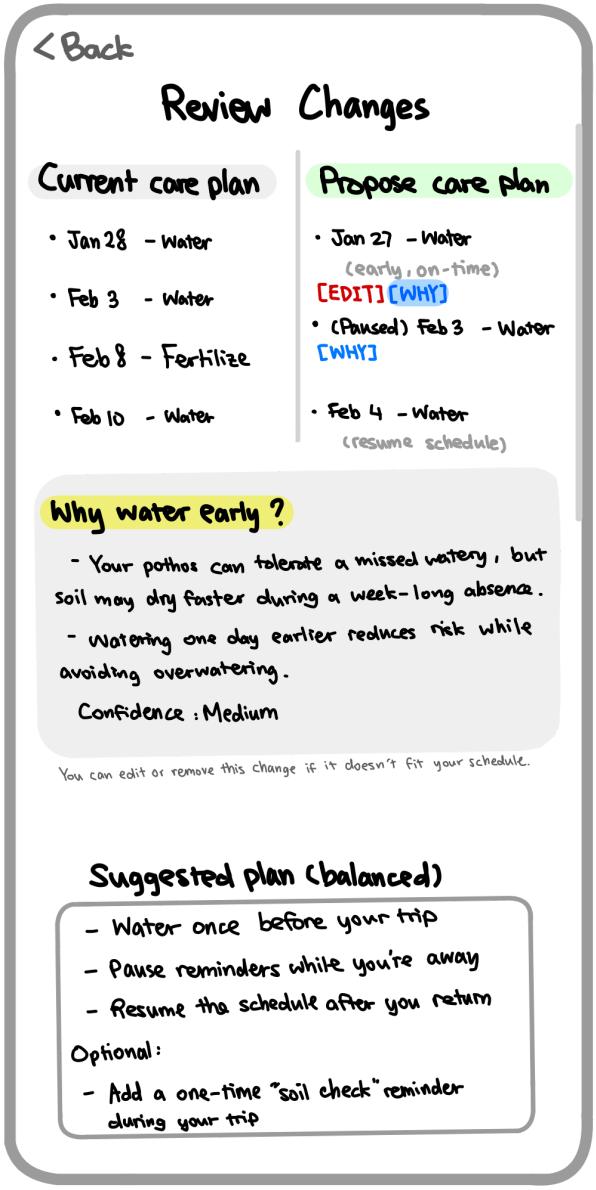


Jamie uses chat to request a short-term change: “I’m away from Jan 28 to Feb 3.” PlantPal asks follow-up questions to collect constraints and information. Then Jamie clicked the “Preview Changes" button.

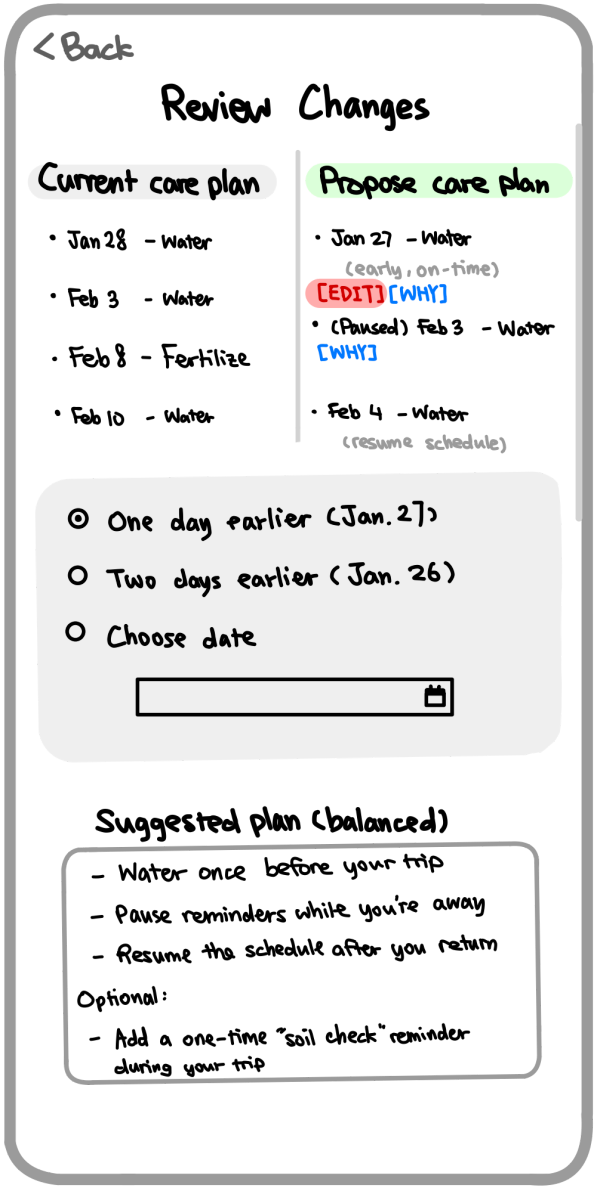
Key changes: the user can describe any situation in chat.



Based on Jamie's input, PlantPal generates a proposed set of concrete schedule changes and presents them for review. The interface shows the current schedule alongside suggested modifications, including moving the Jan 28 watering to Jan 27 as a one-time early action, pausing reminders during the travel period, and resuming the regular schedule on Feb 4. At this stage, the changes are presented as a preview and have not yet been applied.

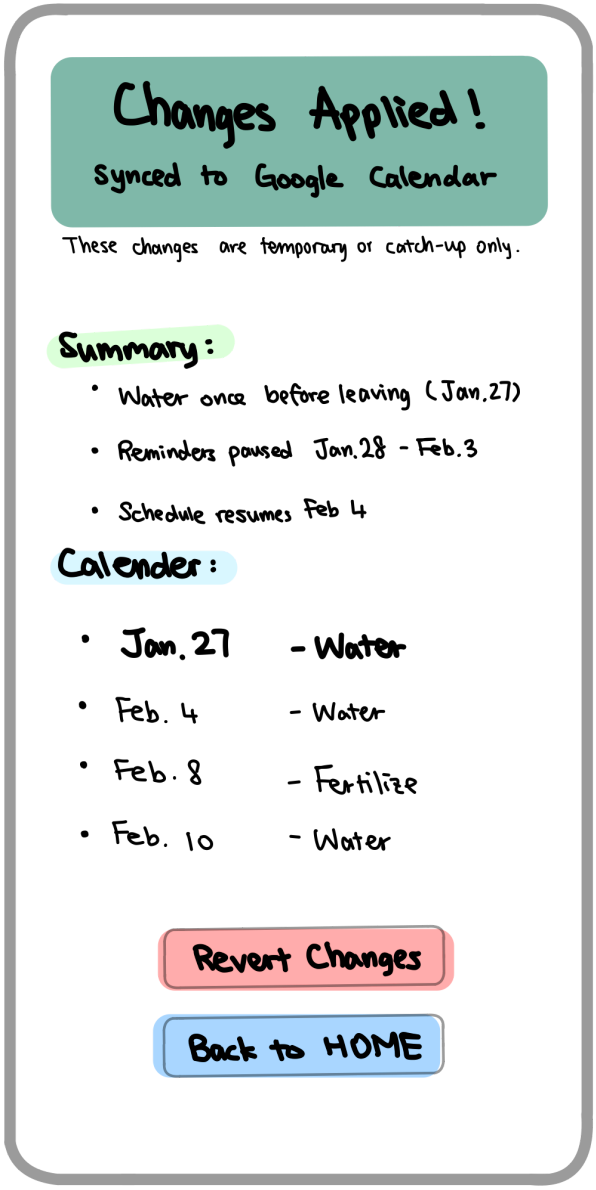


Jamie taps the “Why” button next to the suggested early watering on Jan 27. A side panel expands to explain the system’s reasoning, including plant-specific considerations and the confidence level of the recommendation. This allows Jamie to understand why the system proposed this change without requiring them to read a long explanation by default.



Even though Jamie doesn't edit here, the design explicitly supports optional, lightweight edits inside the preview (e.g., adjusting the day of a one-time task) before anything is applied. The user stays in control without restarting the whole conversation.

Key changes: editing is framed as small plan tweaks (day-level), not hour-level precision.



Jamie confirms the plan after reviewing it. Once the changes are applied, PlantPal adds the updated care events to Jamie's regular Google Calendar. PlantPal does not replace the calendar, but works alongside it by helping Jamie follow through on those events.

Key changes: Google Calendar is used to store the care schedule, while PlantPal focuses on reminding Jamie, allowing snooze, and tracking whether the task was completed.

On the next watering day, when Jamie opens the app, PlantPal immediately prompts a simple “Did you water today?” check-in. Jamie can mark it done, or snooze if they’re busy—so the reminder adapts to the user’s day instead of assuming a perfect schedule. If the day ends without completion, PlantPal treats it as a missed task and automatically rolls the watering forward to the next day as a temporary catch-up adjustment, preventing the plan from silently breaking.

Key changes: the design now shows how reminders are experienced, how snooze works, and how completion is recorded, which was the biggest unclear part in the previous version.

Language Model Support

PlantPal calls the language model when the user describes a disruption or a missed care day. The model receives the user’s message plus a minimized plant-care snapshot pulled by PlantPal, such as upcoming care tasks and user reminder preferences. It doesn’t receive the user’s full calendar. The model returns a structured interpretation of dates and constraints and proposes a small set of schedule adjustments that the UI can preview before syncing. It can also generate an optional short explanation of why each change was suggested and how confident it is. PlantPal only updates Google Calendar after the user applies changes.

Word Count

To support efficient review, ensure you are within word count limits.

Design Summary: 97 words, limit 100
Scenario 1 Task and Story: 100 words, limit 100
Scenario 1 Language Model Support: 96 words, limit 100
Scenario 2 Task and Story: 97 words, limit 100
Scenario 2 Language Model Support: 100 words, limit 100